Anti-Disturbance Proximal Neural Networks for Composite Resource Allocation

Linhua Luan, Shuai Qi, and Sitian Qin

***Abstract*—Composite resource allocation problems with general constraints are studied in this article, which frequently arise in networked systems such as smart grids and multiagent coordination. To address the challenges posed by multivalued differential inclusions resulting from nonsmooth objective functions, two anti-disturbance proximal neural networks are proposed, each tailored to handle structured and unstructured disturbances, respectively. For structured disturbances, the neural network is developed based on the internal model principle, which exploits the underlying structure of the known dynamics. To further improve resilience against unstructured disturbances, another observer-based neural network is designed. The asymptotic convergences of both neural networks are rigorously established using Lyapunov stability theory. Finally, numerical simulations validate the effectiveness and robustness of the proposed neural networks under different types of disturbances.**

***Index Terms*—Anti-disturbance, composite resource allocation, multiagent system, neural network, proximal operators.**

I. Introduction

RESOURCE allocation has become a fundamental problem in modern networked systems due to its crucial role in improving efficiency, reliability, and economic performance across a wide range of applications, including dispatch problems [1], communication networks [2], and microgrid systems [3]. In many practical scenarios, the underlying optimization problems involve composite objective functions consisting of both differentiable terms (e.g., quadratic or smooth utility functions) and nondifferentiable components (e.g., L_1-norms and indicator functions). Such formulations arise naturally from economic dispatch, sparsity-promoting control, and operational constraints, but they also introduce additional analytical and computational challenges [4], [5].

Building on the pioneering work of Hopfield and Tank [6], recurrent neural networks have been widely studied for diverse optimization problems owing to their neurobiological plausibility and hardware implementability [7], [8]. To handle optimization problems where the objective function exhibits a composite structure with both smooth and nonsmooth components, several neural networks were proposed for composite resource allocation. For example, a consensus-based neural network and an ADMM algorithm were proposed in [4] and [9], respectively, for unconstrained composite distributed optimization problems. However, in composite resource allocation problems, there typically exists an inherent resource balance constraint—an equality constraint that couples all node variables. To this end, a proximal nested primal-dual gradient neural network was proposed in [10], and a momentum-based iterative algorithm was discussed in [11] under the case of nonlinearity and latency. Although the recent survey [12] provides a comprehensive review of effective discrete-time distributed neural networks for dealing with resource allocations, it seems dependent on the selection of step size and has limited adaptability to the objective physical limitations in the real environment.

Compared with the discrete-time neural networks mentioned in [12], continuous-time neural networks can more naturally simulate the dynamic evolution process of the system by controlling the dynamic behavior of agents, as demonstrated in recent works [13], [14], [15], [16]. Building on this advantage, continuous-time proximal neural networks have emerged as a powerful tool for optimizing nonsmooth objective functions. In particular, a distributed proximal neural network was developed in [17] to address nonsmooth problems without additional constraints. To extend this framework to second-order multiagent systems, derivative-feedback terms were introduced in [18]. Furthermore, advancements include a projection-based proximal neural network for local set constraints [19], and a multiproximal approach for problems involving both coupling inequalities and local constraints [20]. Notably, by using the properties of proximal operators, distributed proximal neural networks overcome the uncertainty caused by the multivalued differential inclusion in the traditional subgradient-based distributed neural networks [21], [22], [23], [24], and the limitation that smooth approximation neural networks cannot converge to the exact optimal solution of the problem [25], [26], [27]. However, existing research has mainly focused on idealized scenarios and paid insufficient attention to external disturbances, which often make a significant impact on real-world applications.

Disturbances in multiagent systems generally refer to factors that disrupt normal system operations. These can include internal disturbances, such as unknown actuator or sensor faults in fault-tolerant control studies [28], [29], [30], [31], as well as external disturbances, such as payload-induced perturbations that negatively affect quadrotor performance [32]. Such disturbances typically degrade the performance, stability, and reliability of the overall system, thereby motivating the development of anti-disturbance neural network approaches

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for enhanced robustness in practical applications. For structured disturbances with known characteristics, a time-triggered neural network was designed in [33] with discrete communication and gradient measurement. In addition, based on the internal model design, the structured disturbances of second-order multiagent systems were suppressed in [34], and the suppression of mismatched constant disturbance was further achieved by the adaptive control technology and the state integral feedback control. Against a wider range of unstructured disturbances, a distributed optimization neural network was designed in [35] using an observer and distributed decoupling control method. More recently, several works have further advanced the anti-disturbance design. In [36], a composite disturbance observer-based controller was developed for hidden Markov jump systems under replay attacks, improving resilience against multiple disturbances. In [37], the disturbance rejection problem for helicopter systems under multiple random disturbances was addressed by designing a stochastic anti-disturbance controller that integrates feedback linearization, backstepping, and disturbance observer techniques.

Motivated by the presence of various disturbances in practical applications, this article develops a novel distributed proximal neural network framework for solving nonsmooth composite resource allocation problems under both structured and unstructured disturbances. Unlike existing methods that mainly address smooth or disturbance-free settings [38], [39], the proposed approach explicitly integrates proximal operators with neural dynamics to handle nonsmooth objectives and heterogeneous disturbance models in a fully distributed manner, thereby advancing the state of the art in robustness and scalability of distributed optimization.

The specific contributions of this article can be concluded as the following points.

1) We develop a distributed proximal neural network that embeds proximal operators into an exact penalty framework. This design implicitly guarantees feasibility while overcoming the limitations of existing proximal neural networks [17], [18], [19], [20], [40], which are typically subject to simple constraints and are sensitive to initial conditions.
2) Unlike conventional methods that depend on subgradient information [21], [22], [23], [24] or inexact smoothing techniques [25], [26], [27], the proposed algorithm directly characterizes the exact optimal solution through proximal mappings. This leads to improved solution accuracy and enhanced computational efficiency for nonsmooth composite objectives.
3) The proposed neural networks explicitly address both structured and unstructured disturbances in multiagent systems. In particular, their disturbance-suppression capability significantly improves adaptability to complex and uncertain environments compared with existing composite optimization methods [9], [10], [17], [18], [41], [42].

The remainder of this article is structured as follows. Section II provides a brief overview of essential preliminaries, including notations, proximal operators, and concepts of graph theory. The formulation of nonsmooth composite resource allocations is presented in Section III. In Section IV, two penalty-based anti-disturbance distributed proximal neural networks are proposed, and their convergence properties are analyzed. Section V displays the simulation results, and Section VI concludes this article.

II. Preliminaries

This section introduces the notations used throughout this article and reviews fundamental concepts from proximal operator theory and graph theory.

A. Notations

For the set of real numbers $\mathbb{R}$, $\mathbb{R}_{>0}$, and $\mathbb{R}^{n\times d}$ denote, respectively, the set of positive real numbers and the space of real $n \times d$ matrices. The notation $(\cdot)^\top$ represents the transpose operation. For any $x \in \mathbb{R}^d$, $\|x\|_1$ and $\|x\|$ denote the L_1-norm and L_2-norm of x, respectively. $A\otimes B$ represents the Kronecker product for matrices A and B. For any positive integer $n > 1$, $\mathcal{I}_n = \{1, 2, \ldots, n\}$. Let $\mathrm{col}(x_i)$ be a column stack of the vectors x_i, and $\mathrm{diag}(x_i)$ be a diagonal matrix with diagonal entries being x_i. For any $a, d \in \mathbb{R}$, $(a)_d = [a, a, \ldots, a]^\top \in \mathbb{R}^d$, and $I_d \in \mathbb{R}^{d\times d}$ is the identity matrix. $\nabla f(x)$ denotes the gradient of at x for a smooth function $f : \mathbb{R}^d \to \mathbb{R}$. $\mathrm{int}(\mathcal{M})$ denotes the interior of the subset $\mathcal{M} \subseteq \mathbb{R}^d$. $\mathcal{D}(p, \mathcal{M})$ denotes the distance from a point p to the set $\mathcal{M}$, that is,

$$\mathcal{D}(p, \mathcal{M}) = \inf_{x\in\mathcal{M}} \|p - x\|.$$

Let $\Omega \subset \mathbb{R}^d$ be a nonempty closed convex set, then the projection is defined by

$$\mathcal{P}_\Omega(v) = \arg\min_{u\in\Omega} \|v - u\|.$$

B. Proximal Operators

For a nonsmooth function $f : \mathbb{R}^d \to \mathbb{R}$, define ∂f as its subdifferential [43]. If $f(x)$ is convex, $\partial f(x)$ is monotone, i.e., $(g_1 - g_2)^\top(x_1 - x_2) \ge 0$, where $g_1 \in \partial f(x_1)$, $g_2 \in \partial f(x_2)$. For $v \in \mathbb{R}^d$, let $\mathrm{Prox}_f(v) \in \mathbb{R}^d$ be the proximal operator of f at v, that is,

$$\begin{aligned}\inf_{u\in\mathbb{R}^d}&\left(f(u) + \frac{1}{2}\|v - u\|^2\right)\\ &= f\left(\mathrm{Prox}_f(v)\right) + \frac{1}{2}\left\|v - \mathrm{Prox}_f(v)\right\|^2.\end{aligned}$$

The definition of $\mathrm{Prox}_f(v)$ implies that $v - \mathrm{Prox}_f(v) \in \partial f(\mathrm{Prox}_f(v))$, and $\|\mathrm{Prox}_f(u) - \mathrm{Prox}_f(v)\| \le \|u - v\|$ for any $u, v \in \mathbb{R}^d$.

C. Graph Theory

The communication network consisting of N agents is modeled by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \mathcal{I}_N$ denotes the set of agents and $\mathcal{E} \subset \mathcal{V}\times\mathcal{V}$ represents the set of edges. An undirected edge $(i, j) \in \mathcal{E}$ indicates that agent i can exchange information with agent j. A path from agent i to agent j is defined as a sequence of edges (i, i_1), (i_1, i_2), …, (i_k, j). The graph $\mathcal{G}$ is said to be *connected* if there exists a path between every pair of agents. Let $A = [a_{ij}] \in \mathbb{R}^{N\times N}$ be the adjacency matrix associated with $\mathcal{G}$, where $a_{ij} = a_{ji} = 1$ if $(i, j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. We assume that $\mathcal{G}$ has no self-loops, i.e., $a_{ii} = 0$. The Laplacian matrix of $\mathcal{G}$ is denoted by $L_{\mathcal{G}} = [l_{ij}] \in \mathbb{R}^{N\times N}$, where $l_{ii} = \sum_{j=1}^{N} a_{ij}$ for each $i \in \mathcal{V}$ and $l_{ij} = -a_{ij}$ for $i \neq j$. By the definition of $L_{\mathcal{G}}$, for any undirected connected graph $\mathcal{G}$, it follows that $(1)_N^\top L_{\mathcal{G}} = L_{\mathcal{G}}(1)_N = (0)_N$. The eigenvalues of $L_{\mathcal{G}}$ can be ordered as $0 = \chi_1(L_{\mathcal{G}}) < \chi_2(L_{\mathcal{G}}) \leq \chi_3(L_{\mathcal{G}}) \leq \cdots \leq \chi_N(L_{\mathcal{G}})$. Moreover, for any vector u satisfying $(1)_N^\top u = 0$, it holds that $u^\top L_{\mathcal{G}} u \geq \chi_2(L_{\mathcal{G}})\|u\|^2$.

III. Problem Formulation

The purpose of this section is to formulate the composite resource allocations widely used in economic dispatch problems in smart grids (see [44], [45]).

Consider the following composite resource allocation with coupling equality constraint and local set constraints:

$$\begin{aligned} \min \quad & C(x) = \sum_{i=1}^{N} (C_{i1}(x_i) + C_{i2}(x_i)) \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} b_i \qquad x_i \in \Omega_i \end{aligned} \tag{1}$$

where $x_i \in \mathbb{R}^d, x = \mathrm{col}(x_i) \in \mathbb{R}^{Nd}, b_i \in \mathbb{R}^d, \Omega_i \subseteq \mathbb{R}^d$, and Ω_i is a compact convex set, we first make the following common assumptions.

Assumption 1:

1) C_{i1} is smooth and σ_i-strongly convex with $\sigma_i > 0$, ∇C_{i1} is l_i-Lipschitz with $l_i > 0$, C_{i2} is convex and may be nonsmooth.
2) There is $\hat{x}_i \in \mathrm{int}(\Omega_i)$ satisfying $\sum_{i=1}^{N} \hat{x}_i = \sum_{i=1}^{N} b_i$.

Remark 1: The strong convexity and Lipschitz continuity conditions in Assumption 1 are often satisfied or can be reasonably approximated in many practical scenarios, such as power systems and economic dispatch problems [46], [47], where quadratic cost functions and regularized utility functions typically exhibit these properties. From a theoretical perspective, the strong convexity of the objective functions guarantees the uniqueness of the optimal solution, while condition 2) implies the existence of feasible solutions. This assumption ensures the solvability of composite resource allocation (1), and it has been widely used in the neural network design for distributed recourse allocations, such as continuous-time neural networks [19], [38], [39], discrete-time neural network [48].

The composite resource allocation problem (1) can be viewed as a nonsmooth distributed optimization problem, as investigated in [49]. Inspired by the exact penalty technique introduced in [49], we assert that there exists a $\theta^* \in \mathbb{R}_{>0}$ such that, for any $\theta > \theta^*$, the optimal solution of problem (1) is equivalent to that of the following problem:

$$\begin{aligned} \min \quad & \sum_{i=1}^{N} [(C_i(x_i) + \theta\mathcal{D}_i(x_i)] \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = \sum_{i=1}^{N} b_i \end{aligned} \tag{2}$$

where $\mathcal{D}_i(x_i) = \mathcal{D}(x_i, \Omega_i)$.

For resource allocation (2), the Lagrange function is given by

$$\mathcal{L}(x, \lambda) = \sum_{i=1}^{N} [C_i(x_i) + \theta\mathcal{D}_i(x_i)] + \lambda^\top \left(\sum_{i=1}^{N} x_i - \sum_{i=1}^{N} b_i \right)$$

where $\lambda \in \mathbb{R}^d$ is the Lagrange multiplier. On the basis of the KKT condition, it is trivial to obtain the following lemma.

Lemma 1 ([50]): From Assumption 1, $x^* = \mathrm{col}(x_i^*)$ is the optimal solution of resource allocation (2) if and only if there is $\lambda^* \in \mathbb{R}^d$ satisfying

$$\begin{aligned} & (0)_{Nd} \in \nabla C_{i1}\left(x_i^*\right) + \partial C_{i2}\left(x_i^*\right) + \theta\partial\mathcal{D}_i\left(x_i^*\right) + \lambda^* \\ & \sum_{i=1}^{N} x_i^* = \sum_{i=1}^{N} b_i \end{aligned} \tag{3}$$

for all $i \in \mathcal{I}_N$.

Remark 2: Inspired by the operator separation technique [18], we introduce auxiliary variables $y_i^*, z_i^* \in \mathbb{R}^d$ to factor the first term in (3) as

$$\begin{aligned} & (0)_{Nd} = \nabla C_{i1}\left(x_i^*\right) + \gamma_1 y_i^* + \theta\gamma_2 z_i^* + \lambda^* \\ & \gamma_1 y_i^* \in \partial C_{i2}\left(x_i^*\right), \ \gamma_2 z_i^* \in \partial\mathcal{D}_i\left(x_i^*\right). \end{aligned} \tag{4}$$

According to the definition of proximal operators, (4) can be converted to

$$\begin{aligned} (0)_{Nd} &= -\nabla C_{i1}\left(x_i^*\right) - \gamma_1 y_i^* - \theta\gamma_2 z_i^* - \lambda^* \\ x_i^* &= \mathrm{Prox}_{C_{i2}}\left(x_i^* + \gamma_1 y_i^*\right) \\ x_i^* &= \mathrm{Prox}_{\mathcal{D}_i}\left(x_i^* + \gamma_2 z_i^*\right). \end{aligned} \tag{5}$$

Obviously, auxiliary variables are used to estimate the subgradient of the nonsmooth term C_{i2} and the distance function $\mathcal{D}_i$, avoiding the multivalued operation in the form of differential inclusion.

IV. Main Results

This section aims to develop distributed neural networks that enable multiagent systems to address the composite resource allocation (1) under structured and unstructured disturbances.

To determine the optimal solution, we examine a specific category of multiagent systems comprising N agents that communicate over an undirected and connected graph $\mathcal{G}$, where the agent i's dynamics are characterized by

$$\dot{x}_i = u_i + \varepsilon_i \tag{6}$$

where x_i, u_i, and ε_i denote the state, control input, and disturbances, respectively.

In practical applications, disturbances ε_i can arise from various sources, such as environmental fluctuations, sensor noise, and model uncertainties (see [33], [34], [35]). These disturbances can generally be classified into two types: structured disturbances with known characteristics, which can be modeled and compensated using internal models, and unstructured disturbances with unknown dynamics, which require adaptive estimation techniques for effective mitigation. To proceed, we analyze both cases and propose corresponding anti-disturbance strategies to ensure the effectiveness of the proposed proximal neural networks.

A. Structured Disturbances With Known Characteristics

We first consider the case where each agent is subject to a structured disturbance with known dynamics. In particular, the disturbance is modeled as

$$\begin{aligned}\dot{e}_i &= A_i e_i\\ \varepsilon_i &= B_i e_i\end{aligned}\tag{7}$$

where $e_i \in \mathbb{R}^{d_i}$, $A_i \in \mathbb{R}^{d_i\times d_i}$, and $B_i \in \mathbb{R}^{d\times d_i}$ are known characteristics.

Assumption 2: Disturbances system (7) is neutrally stable and observable.

Remark 3: Assumption 2 is widely used in the design of disturbances. As described in [51], [52], and [53], the eigenroots of the system matrix A_i are all on the virtual axis, which makes it possible to suppress the disturbance ε_i effectively.

For agent i, the control input is designed as

$$\begin{aligned}u_i = &-\nabla C_{i1}(x_i) - \gamma_1 y_i - \theta\gamma_2 z_i - \theta\gamma_2\lambda_i - 2\dot{y}_i - 2\dot{z}_i\\ &+ B_i(K_i x_i - \delta_i)\end{aligned}\tag{8}$$

where δ_i is an internal compensator state, $K_i \in \mathbb{R}^{d_i\times d}$ is feedback gain such that $A_i + K_iB_i$ is Hurwitz, the penalty parameter θ primarily governs the robustness and accuracy with respect to constraints, and the scaled parameters $\gamma_1 > 0$ and $\gamma_2 > 0$ control the adaptability and convergence behavior of the algorithmic response.

The auxiliary dynamics are given by

$$\begin{aligned}\dot{y}_i &= x_i - \mathrm{Prox}_{C_{i2}}(x_i + \gamma_1 y_i)\\ \dot{z}_i &= \theta x_i - \theta\mathrm{Prox}_{\mathcal{D}_i}(x_i + \gamma_2 z_i)\\ \dot{\lambda}_i &= \theta\gamma_2(x_i - b_i) - \nabla C_{i1}(x_i) - \gamma_1 y_i - \theta\gamma_2 z_i - \theta\gamma_2\lambda_i\\ &\quad - c\sum_{j=1}^{N} a_{ij}(\lambda_i - \lambda_j) - c\sum_{j=1}^{N} a_{ij}(w_i - w_j)\\ &\quad + B_i(K_i x_i + e_i - \delta_i)\\ \dot{w}_i &= c\sum_{j=1}^{N} a_{ij}(\lambda_i - \lambda_j) - B_i(Kx_i + e_i - \delta_i)\\ \dot{\delta}_i &= -A_i(K_i x_i - \delta_i) + K_i(u_i - B_i(K_i x_i - \delta_i))\end{aligned}$$

where y_i and z_i are responsible for handling the local proximal operations associated with the objective function and the local constraints, respectively. The variable λ_i coordinates the dual information corresponding to the coupling constraints, while w_i ensures the consensus propagation of the local estimates λ_i across the communication network. Moreover, δ_i acts as a disturbance observer and compensator, reconstructing the structured disturbance using the known matrices A_i, B_i, and feedback through K_i. By defining the error variable $r_i = K_i x_i + e_i - \delta_i$, the linear error dynamics

$$\dot{r}_i = (A_i + K_iB_i)r_i$$

are obtained, which guarantee exponential convergence $r_i \to 0$ because $A_i + K_iB_i$ is Hurwitz. In summary, y, z, λ, and w form the core distributed mechanism for handling composite objectives and coupling constraints, while δ_i and r_i enable effective estimation and rejection of structured disturbances.

Thus, the closed-loop form of the distributed neural network (6)–(8) is

$$\begin{aligned}\dot{x} &= -\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} + Br\\ \dot{y} &= x - \mathrm{Prox}_{C_2}(x + \gamma_1 y)\\ \dot{z} &= \theta x - \theta\mathrm{Prox}_{\mathcal{D}}(x + \gamma_2 z)\\ \dot{\lambda} &= \theta\gamma_2(x - b) - \nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda\\ &\quad - cL\lambda - cLw + Br\\ \dot{w} &= cL\lambda - Br\\ \dot{r} &= (A + KB)r\end{aligned}\tag{9}$$

where $y = \mathrm{col}(y_i)$, $z = \mathrm{col}(z_i)$, $\lambda = \mathrm{col}(\lambda_i)$, $w = \mathrm{col}(w_i)$, $r = \mathrm{col}(r_i)$, $\nabla C_1(x) = \mathrm{col}(\nabla C_{i1}(x_i))$, $\mathrm{Prox}_{\mathcal{D}}(\omega) = \mathrm{col}(\mathrm{Prox}_{\mathcal{D}_i}(\omega_i))$, $\mathrm{Prox}_{C_2}(\omega) = \mathrm{col}(\mathrm{Prox}_{C_{i2}}(\omega_i))$ $b = \mathrm{col}(b_i)$, $L = L_{\mathcal{G}} \otimes I_d$, $A = \mathrm{diag}(A_i)$, $B = \mathrm{diag}(B_i)$, and $K = \mathrm{diag}(K_i)$.

Theorem 1: Under Assumptions 1 and 2, the equilibrium point $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, r^*)$ of neural network (9) implies that x^* is the optimal solution of resource allocation (2). Furthermore, if x^* is the optimal solution of resource allocation (2), there is $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, r^*)$ to be the equilibrium point of neural network (9).

Proof: By the nature of the equilibrium point, we have

$$(0)_{Nd} = -\nabla C_1(x^*) - \gamma_1 y^* - \theta\gamma_2 z^* - \theta\gamma_2\lambda^* + Br^* \tag{10a}$$
$$x^* = \mathrm{Prox}_{C_2}(x^* + \gamma_1 y^*) \tag{10b}$$
$$x^* = \mathrm{Prox}_{\mathcal{D}}(x^* + \gamma_2 z^*) \tag{10c}$$
$$(0)_{Nd} = \theta\gamma_2(x^* - b) - cL\lambda^* - cLw^* \tag{10d}$$
$$(0)_{Nd} = cL\lambda^* - Br^* \tag{10e}$$
$$(0)_D = (A + KB)r^* \tag{10f}$$

where $D = \sum_{i=1}^{N} d_i$.

Because $A + KB$ is Hurwitz matrix, (10f) can deduce $r^* = (0)_D$, so (10a) and (10e) can be reduced to

$$(0)_{Nd} = \nabla C_1(x^*) + \gamma_1 y^* + \theta\gamma_2 z^* + \theta\gamma_2\lambda^* \tag{11}$$

and

$$(0)_{Nd} = cL\lambda^*. \tag{12}$$

From Assumption 1, (11) and (12) can deduce that there exists $\lambda_0 \in \mathbb{R}^d$, such that $\lambda^* = 1/(\theta\gamma_2)((1)_N \otimes \lambda_0)$, and for any i

$$\nabla C_{i1}(x_i^*) + \gamma_1 y_i^* + \theta\gamma_2 z_i^* + \lambda_0 = (0)_d.$$

Combine with (10b), (10c), and properties of the proximal operators, it has $\gamma_1 y_i^* \in \partial C_{i1}(x_i^*)$ and $\gamma_2 z_i^* \in \partial\mathcal{D}_i(x_i^*)$. Then, one holds

$$(0)_d \in \nabla C_{i1}(x_i^*) + \gamma_1\partial C_{i2}(x_i^*) + \theta\gamma_2\partial\mathcal{D}_i(x_i^*) + \lambda_0.$$

Multiplying by $(1)_N^\top$ from the left, (10d) becomes $\sum_{i=1}^{N} x_i^* = \sum_{i=1}^{N} b_i$. Therefore, it follows from Lemma 1 that x^* is the optimal solution of resource allocation (2).

In addition, if x^* is the optimal solution of resource allocation (2), according to Lemma 1 and (5), $y^* = \mathrm{col}(\hat{y}_i^*)$, $z^* = \mathrm{col}(\hat{z}_i^*)$, $\lambda^* = 1/(\theta\gamma_2)(1)_N \otimes \hat{\lambda}^*$, and $r^* = 0$ can make (10a)–(10c), (10e), and (10f) hold, where $\hat{y}_i^*$, $\hat{z}_i^*$, and $\hat{\lambda}_i^*$ satisfy (3). Since $(1)_N^\top(x^* - b) = \sum_{i=1}^{N} x_i^* - \sum_{i=1}^{N} b_i = 0$, it has $x^* - b \in \mathrm{Range}(L)$, that is, there is $\hat{w}_i^* \in \mathbb{R}^{Nd}$ such that $x^* - b = L\hat{w}_i^*$. Then, (10d) holds by taking $w^* = (\theta\gamma_2)/c\hat{w}_i^*$.

To sum up, we can find $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, r^*)$ to be the equilibrium point of neural network (9).

Theorem 2: Under Assumptions 1 and 2, if $\gamma_1 = \theta\gamma_2$, $\gamma_2 > (2l^2)/(\theta\sigma)$, and $c > \max\{(\|L\|^2(3\theta\gamma_2 + l))/(2\chi_2(L)), (1 + \|L\|^2(1 + 3\theta\gamma_2 + l))/(2\chi_2^2(L))\}$ with $\sigma = \min\{\sigma_i\}$, $l = \max\{l_i\}$, then the state solution $\mathrm{col}(x(t), y(t), z(t), \lambda(t), w(t), r(t))$ generated by neural network (9) converges to equilibrium point $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, r^*)$.

Proof: Define the Lyapunov function $V = V_1 + V_2$ with

$$V_1 = \begin{pmatrix}\bar{x}\\ \bar{y}\\ \bar{z}\end{pmatrix}^\top \left[\begin{pmatrix}\frac{\phi_2}{2} & \frac{\phi_2}{2} & \frac{\phi_2}{2}\\ \frac{\phi_2}{2} & \frac{\gamma_1\phi_2}{2}+\phi_2 & \phi_2\\ \frac{\phi_2}{2} & \phi_2 & \frac{\gamma_2\phi_2}{2}+\phi_2\end{pmatrix} \otimes I_{Nd}\right] \begin{pmatrix}\bar{x}\\ \bar{y}\\ \bar{z}\end{pmatrix} + \frac{\phi_2}{2}\left\|\bar{\lambda}\right\|^2 + \frac{\phi_1}{2}\left\|L^{\frac{1}{2}}\left(\bar{\lambda}+\bar{w}\right)\right\|^2 + \frac{\phi_2}{2}\|\bar{w}\|^2 + \frac{\phi_1}{2}\left\|L^{\frac{1}{2}}\bar{w}\right\|^2$$

$$V_2 = \bar{r}^\top Q\bar{r}$$

where $\bar{x} = x - x^*$, $\bar{y} = y - y^*$, $\bar{z} = z - z^*$, $\bar{\lambda} = \lambda - \lambda^*$, $\bar{w} = w - w^*$, and $\bar{r} = r - r^*$. Obviously, V is positive define about $\bar{\xi} \triangleq \mathrm{col}(\bar{x}, \bar{y}, \bar{z}, \bar{\lambda}, \bar{w}, \bar{r})$, and $V(\bar{\xi}) = 0$ if and only if $\bar{\xi} = 0_{5Nd+D}$. By calculating, there is

$$\begin{aligned}\dot{V} = &-\phi_2 (x - x^*)^\top (\nabla C_1(x) - \nabla C_1(x^*)) - \phi_2\gamma_1\bar{x}^\top\bar{y}\\
&- \phi_2\theta\gamma_2\bar{x}^\top\bar{z} - \phi_2\theta\gamma_2\bar{x}^\top\bar{\lambda} - 2\phi_2\bar{x}^\top\dot{y} - 2\phi_2\bar{x}^\top\dot{z}\\
&+ \phi_2\bar{x}^\top B\bar{r} + \phi_2\gamma_1\bar{y}^\top\dot{y} + 2\phi_2\bar{y}^\top\dot{y} + \phi_2\bar{x}^\top\dot{y}\\
&+ \phi_2\bar{y}^\top\left[-\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} + Br\right]\\
&+ 2\phi_2\bar{y}^\top\dot{z} + \phi_2\gamma_2\bar{z}^\top\dot{z} + 2\phi_2\bar{z}^\top\dot{z} + \phi_2\bar{x}^\top\dot{z}\\
&+ \phi_2\bar{z}^\top\left[-\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} + Br\right]\\
&+ 2\phi_2\bar{z}^\top\dot{y} - c\phi_2\bar{\lambda}^\top L\bar{\lambda} - c\phi_2\bar{\lambda}^\top L\bar{w} + \phi_2\theta\gamma_2\bar{\lambda}^\top\bar{x}\\
&- \phi_2\bar{\lambda}^\top(\nabla C_1(x) - \nabla C_1(x^*)) - \phi_2\gamma_1\bar{\lambda}^\top\bar{y} - \phi_2\theta\gamma_2\bar{\lambda}^\top\bar{z}\\
&- \phi_2\theta\gamma_2\bar{\lambda}^\top\bar{\lambda} + \phi_2\bar{\lambda}^\top B\bar{r} + c\phi_2\bar{w}^\top L\bar{\lambda} - \phi_2\bar{w}^\top B\bar{r}\\
&+ \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{x} - \phi_1\bar{\lambda}^\top L(\nabla C_1(x) - \nabla C_1(x^*))\\
&- \phi_1\gamma_1\bar{\lambda}^\top L\bar{y} - \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{z} - \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{\lambda}\\
&- c\phi_1\bar{\lambda}^\top L^2\bar{w} + \phi_1\theta\gamma_2\bar{w}^\top L\bar{x}\\
&- \phi_1\bar{w}^\top L(\nabla C_1(x) - \nabla C_1(x^*)) - \phi_1\gamma_1\bar{w}^\top L\bar{y}\\
&- \phi_1\theta\gamma_2\bar{w}^\top L\bar{z} - \phi_1\theta\gamma_2\bar{w}^\top L\bar{\lambda} - c\phi_1\bar{w}^\top L^2\bar{w}\\
&+ c\phi_1\bar{w}^\top L^2\bar{\lambda} - \phi_1\bar{w}^\top LB\bar{r} + \bar{r}^\top\left(Q + Q^\top\right)\dot{r}.\end{aligned}$$

From dynamics of y and z in the neural network (9), one can get $x - \dot{y} = \mathrm{Prox}_{C_2}(x + \gamma_1 y)$, $x - 1/\theta\dot{z} = \mathrm{Prox}_{\mathcal{D}}(x + \gamma_2 z)$, and $x^* = \mathrm{Prox}_{C_2}(x^* + \gamma_1 y^*) = \mathrm{Prox}_{\mathcal{D}}(x^* + \gamma_2 z^*)$. According to properties of proximal operators, we have

$$x + \gamma_1 y - (x - \dot{y}) = \gamma_1 y + \dot{y} \in \partial C_2(x - \dot{y})$$

$$x + \gamma_2 z - \left(x - \frac{1}{\theta}\dot{z}\right) = \gamma_2 z + \frac{1}{\theta}\dot{z} \in \partial\mathcal{D}\left(x - \frac{1}{\theta}\dot{z}\right)$$

$\gamma_1 y^* \in \partial C_2(x^*)$, and $\gamma_2 z^* \in \partial\mathcal{D}(x^*)$. Combining with the convexity of $C_2(x)$ and $\mathcal{D}(x)$, it holds that $(\gamma_1 y + \dot{y} - \gamma_1 y^*)^\top(x - \dot{y} - x^*) \geq 0$ and $(\gamma_2 z + 1/\theta\dot{z} - \gamma_2 z^*)^\top(x - 1/\theta\dot{z} - x^*) \geq 0$, which means that $(\gamma_1\bar{y} + \dot{y})^\top(\bar{x} - \dot{y}) = \gamma_1\bar{y}^\top\bar{x} + \dot{y}^\top\bar{x} - \gamma_1\bar{y}^\top\dot{y} - \|\dot{y}\|^2 \geq 0$ and $(\gamma_2\bar{z} + 1/\theta\dot{z})^\top(\bar{x} - 1/\theta\dot{z}) = \gamma_2\bar{z}^\top\bar{x} + 1/\theta\dot{z}^\top\bar{x} - \gamma_2/\theta\bar{z}^\top\dot{z} - 1/\theta^2\|\dot{z}\|^2 \geq 0$. Therefore, we have

$$(\gamma_1\bar{y} - \bar{x})^\top\dot{y} \leq \gamma_1\bar{y}^\top\bar{x} - \|\dot{y}\|^2 \tag{13}$$

and

$$(\gamma_2\bar{z} - \bar{x})^\top\dot{z} \leq \theta\gamma_2\bar{z}^\top\bar{x} - \frac{1}{\theta}\|\dot{z}\|^2. \tag{14}$$

According to [54] and $A + KB$ is Hurwitz matrix, there exits a positive-definite matrix Q such that

$$(A + KB)^\top Q + Q(A + KB) = -\eta\|B\|^2 I_D \tag{15}$$

where $\eta = \phi_2/2\sigma + \phi_2/(\theta\gamma_2) + (\phi_2^2)/(2\phi_1) + (\phi_1)/2 + 1$ with $\phi_1, \phi_2 > 0$. Hence, the strong convexity of $C_1(x)$ and (15) follows that:

$$\begin{aligned}\dot{V} \leq &-\phi_2\sigma\|\bar{x}\|^2 - \phi_2\|\dot{y}\|^2 - \frac{\phi_2}{\theta}\|\dot{z}\|^2 + \phi_2\bar{x}^\top B\bar{r} + \phi_2\bar{y}^\top B\bar{r}\\
&+ \phi_2\bar{z}^\top B\bar{r} + \phi_2\bar{\lambda}^\top B\bar{r} - \phi_2\bar{w}^\top B\bar{r} - \phi_1\bar{w}^\top LB\bar{r}\\
&- \eta\|B\|^2\|\bar{r}\|^2 - \phi_2\left(\bar{y} + \bar{z} + \bar{\lambda}\right)^\top(\nabla C_1(x) - \nabla C_1(x^*))\\
&- \phi_2\gamma_1\bar{y}^\top\bar{y} - \phi_2\theta\gamma_2\bar{y}^\top\bar{z} - \phi_2\theta\gamma_2\bar{y}^\top\bar{\lambda} - \phi_2\gamma_1\bar{z}^\top\bar{y}\\
&- \phi_2\theta\gamma_2\bar{z}^\top\bar{z} - \phi_2\theta\gamma_2\bar{z}^\top\bar{\lambda} - \phi_2\gamma_1\bar{\lambda}^\top\bar{y} - \phi_2\theta\gamma_2\bar{\lambda}^\top\bar{z}\\
&- \phi_2\theta\gamma_2\bar{\lambda}^\top\bar{\lambda} - c\phi_2\bar{\lambda}^\top L\bar{\lambda} - \phi_1\bar{\lambda}^\top L(\nabla C_1(x) - \nabla C_1(x^*))\\
&+ \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{x} - \phi_1\gamma_1\bar{\lambda}^\top L\bar{y} - \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{z}\\
&- \phi_1\theta\gamma_2\bar{\lambda}^\top L\bar{\lambda} + \phi_1\theta\gamma_2\bar{w}^\top L\bar{x}\\
&- \phi_1\bar{w}^\top L(\nabla C_1(x) - \nabla C_1(x^*)) - \phi_1\gamma_1\bar{w}^\top L\bar{y}\\
&- \phi_1\theta\gamma_2\bar{w}^\top L\bar{z} - \phi_1\theta\gamma_2\bar{w}^\top L\bar{\lambda} - c\phi_1\bar{w}^\top L^2\bar{w}.\end{aligned}$$

Due to the fact that the eigenvalues $0 < \chi_2(L) \leq \cdots \leq \chi_N(L)$ of L and the eigenvalues $0 < \chi_2^2(L) \leq \cdots \leq \chi_N^2(L)$ correspond to the same eigenvectors, denote $\mathrm{col}(\lambda^{\parallel}, \lambda^{\perp}) = T\bar{\lambda}$ with the orthogonal matrix $T = [R_1, R_2]^\top \otimes I_d$, $R_1 = 1/((N)^{1/2})(1)_N$, one holds $L\lambda^{\parallel} = (0)_{Nd}$. Therefore, letting $\gamma_1 = \theta\gamma_2$, we have

$$\begin{aligned}\dot{V} \leq &-\phi_2\sigma\|\bar{x}\|^2 - \phi_2\|\dot{y}\|^2 - \frac{\phi_2}{\theta}\|\dot{z}\|^2 + \phi_2\bar{x}^\top B\bar{r}\\
&+ \phi_2\left(\bar{y} + \bar{z} + \bar{\lambda}\right)^\top B\bar{r} - \phi_2\bar{w}^\top B\bar{r} - \phi_1\bar{w}^\top LB\bar{r}\\
&- \eta\|B\|^2\|\bar{r}\|^2 + \phi_2 l\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|\|\bar{x}\|\\
&- \phi_2\theta\gamma_2\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2 - c\phi_2\lambda_2(L)\left\|\lambda^{\perp}\right\|^2\\
&+ \phi_1(\theta\gamma_2 + l)\left\|\lambda^{\perp}\right\|\|L\|\|\bar{x}\| + \phi_1(\theta\gamma_2 + l)\|\bar{w}\|\|L\|\|\bar{x}\|\\
&+ \phi_1\theta\gamma_2\left\|\lambda^{\perp}\right\|\|L\|\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|\\
&+ \phi_1\theta\gamma_2\|\bar{w}\|\|L\|\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\| - c\phi_1\chi_2^2(L)\|\bar{w}\|^2.\end{aligned} \tag{16}$$

From Young's inequality, it follows:

$$\begin{aligned}&\phi_2\bar{x}^\top B\bar{r} + \phi_2\left(\bar{y} + \bar{z} + \bar{\lambda}\right)^\top B\bar{r} - \phi_2\bar{w}^\top B\bar{r} - \phi_1\bar{w}^\top LB\bar{r}\\
&\leq \frac{\phi_2\sigma}{2}\|\bar{x}\|^2 + \frac{\phi_2\theta\gamma_2}{4}\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2 + \frac{\phi_1\left(1 + \|L\|^2\right)}{2}\|\bar{w}\|^2\\
&\quad + \left(\frac{\phi_2}{2\sigma} + \frac{\phi_2}{\theta\gamma_2} + \frac{\phi_2^2}{2\phi_1} + \frac{\phi_1}{2}\right)\|B\|^2\|\bar{r}\|^2\end{aligned}$$

$$\phi_2 l\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|\|\bar{x}\| \leq \frac{\phi_2\theta\gamma_2}{4}\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2 + \frac{\phi_2 l^2}{\theta\gamma_2}\|\bar{x}\|^2$$

$$\left\|\lambda^{\perp}\right\|\|L\|\|\bar{x}\| \leq \frac{\|L\|^2}{2}\left\|\lambda^{\perp}\right\|^2 + \frac{1}{2}\|\bar{x}\|^2$$

$$\left\|\lambda^{\perp}\right\|\|L\|\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\| \leq \|L\|^2\left\|\lambda^{\perp}\right\|^2 + \frac{1}{4}\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2$$

$$\|\bar{w}\|\|L\|\|\bar{x}\| \leq \frac{\|L\|^2}{2}\|\bar{w}\|^2 + \frac{1}{2}\|\bar{x}\|^2$$

$$\|\bar{w}\|\|L\|\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\| \leq \|L\|^2\|\bar{w}\|^2 + \frac{1}{4}\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2.$$

Then, substituting the above inequality into (16) gives us

$$\begin{aligned}\dot{V} \leq & -\frac{\phi_2\sigma}{2}\|\bar{x}\|^2 - \phi_2\|\dot{y}\|^2 - \frac{\phi_2}{\theta}\|\dot{z}\|^2 - \|B\|^2\|\bar{r}\|^2 \\ & + \frac{\phi_2\theta\gamma_2}{4}\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 + \frac{\phi_1}{2}\|\bar{w}\|^2 + \frac{\phi_1\|L\|^2}{2}\|\bar{w}\|^2 \\ & - \phi_2\theta\gamma_2\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 + \frac{\phi_2\theta\gamma_2}{4}\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 \\ & + \frac{\phi_2 l^2}{\theta\gamma_2}\|\bar{x}\|^2 + \frac{\phi_1\theta\gamma_2\|L\|^2}{2}\left\|\lambda^{\perp}\right\|^2 \\ & + \frac{\phi_1\theta\gamma_2}{2}\|\bar{x}\| + \frac{\phi_1 l\|L\|^2}{2}\left\|\lambda^{\perp}\right\|^2 + \frac{\phi_1 l}{2}\|\bar{x}\|^2 \\ & + \phi_1\theta\gamma_2\|L\|^2\left\|\lambda^{\perp}\right\|^2 + \frac{\phi_1\theta\gamma_2}{4}\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 \\ & + \frac{\phi_1\theta\gamma_2\|L\|^2}{2}\|\bar{w}\|^2 + \frac{\phi_1\theta\gamma_2}{2}\|\bar{x}\|^2 \\ & + \frac{\phi_1 l\|L\|^2}{2}\|\bar{w}\|^2 + \frac{\phi_1 l}{2}\|\bar{x}\|^2 \\ & + \phi_1\theta\gamma_2\|L\|^2\|\bar{w}\|^2 + \frac{\phi_1\theta\gamma_2}{4}\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 \\ & - c\phi_1\chi_2^2(L)\|\bar{w}\|^2 - c\phi_2\chi_2(L)\left\|\lambda^{\perp}\right\|^2 \\ = & -k_1\|\bar{x}\|^2 - \phi_2\|\dot{y}\|^2 - \frac{\phi_2}{\theta}\|\dot{z}\|^2 - \|B\|^2\|\bar{r}\|^2 \\ & - k_2\left\|\bar{y}+\bar{z}+\bar{\lambda}\right\|^2 - k_3\left\|\lambda^{\perp}\right\|^2 - k_4\|\bar{w}\|^2 \end{aligned} \tag{17}$$

where $k_1 = (\phi_2\sigma)/2 - (\phi_2 l^2)/(\theta\gamma_2) - \phi_1\theta\gamma_2 - \phi_1 l$, $k_2 = (\phi_2\theta\gamma_2)/2 - (\phi_1\theta\gamma_2)/2$, $k_3 = c\phi_2\chi_2(L) - (3\phi_1\theta\gamma_2\|L\|^2)/2 - (\phi_1 l\|L\|^2)/2$, $k_4 = c\phi_1\chi_2^2(L) - (\phi_1)/2 - (\phi_1\|L\|^2)/2 - (3\phi_1\theta\gamma_2\|L\|^2)/2 - (\phi_1 l\|L\|^2)/2$.

It is trivial to obtain $k_i > 0 (i \in \mathcal{I}_4)$ when parameters meet $\phi_1 < \min\{1, (\sigma/2 - l^2/(\theta\gamma_2))/(\theta\gamma_2 + l)\}\phi_2$, $\phi_2 > 0$, $\gamma_2 > 2l^2/\theta\sigma$, and $c > \max\{(\|L\|^2(3\theta\gamma_2 + l))/(2\chi_2(L)), (1 + \|L\|^2(1 + 3\theta\gamma_2 + l))/(2\chi_2^2(L))\}$. Therefore, from LaSalle's invariance principle, (17) means that the state solution generated by neural network (9) asymptotically converges to the equilibrium point.

Corollary 1: The composite resource allocation (1) can be effectively solved by neural network (9) in the multiagent system (6) with structured disturbance (7).

Proof: Combining with Theorems 1 and 2, there exists $\theta^* \in \mathbb{R}_{>0}$, when $\theta > \theta^*$, the state $x(t)$ generated by neural network (9) converges asymptotically to the optimal solution of the composite resource allocation (1).

Remark 4: In practical implementations, the gain parameters may deviate from their nominal values due to modeling uncertainties or tuning inaccuracies. Nevertheless, the stability of the closed-loop system can still be ensured if such variations remain within prescribed bounds. In particular, as shown in Theorem 2, the Lyapunov function $\dot{V}$ remains negative definite provided that the parameters satisfy the derived selection condition. This implies that small bounded perturbations of the parameters do not affect the asymptotic convergence of the trajectories. Moreover, the passivity-based structure of the proposed neural dynamics contributes to additional robustness margins, preventing instability even under slight mismatches.

Remark 5: Compared with the distributed optimization problems in [52] and [53], this work simultaneously considers coupling equality constraints and external disturbances, significantly expanding the scope of applicability. Notably, neural network (9) uses proximal operators to handle local set constraints, enhancing its capability to manage complex constraint structures.

In contrast to subgradient-based methods that rely on multivalued differential inclusions [22], [23], [24], the proposed approach ensures higher numerical precision and avoids convergence ambiguities. Furthermore, unlike smoothing approximation techniques [25], [26], [27], which may introduce approximation errors and parameter tuning difficulties, the use of proximal operators preserves the original problem structure without sacrificing accuracy. This makes the adopted technique particularly suitable for high-precision distributed optimization under complex constraints.

B. Unstructured Disturbances With Unknown Dynamics

Section III-A proposed a proximal neural network for multiagent systems subject to structured disturbances. In practical applications, such as power grid systems, unstructured disturbances are more common and often more challenging to mitigate. Accordingly, in this section, we design the control protocol u_i for the multiagent system (6) in the presence of unstructured disturbances ε_i.

It is commonly assumed that the variation rate of the unstructured disturbance is bounded, that is, there is ϱ_i satisfying $\|\dot{\varepsilon}_i(t)\| \leq \varrho_i$ for each $i \in \mathcal{I}_N$. To this end, the following observer is proposed to suppress the unstructured disturbances:

$$\begin{aligned}\breve{\varepsilon}_i &= x_i - \xi_i \\ \dot{\xi}_i &= u_i + \breve{\varepsilon}_i - \varrho_i \operatorname{sign}(\varepsilon_i - \breve{\varepsilon}_i).\end{aligned} \tag{18}$$

In this case, the control protocol of agent i is designed as

$$u_i = -\nabla C_{i1}(x_i) - \gamma_1 y_i - \theta\gamma_2 z_i - \theta\gamma_2\lambda_i - 2\dot{y}_i - 2\dot{z}_i - \breve{\varepsilon}_i \tag{19}$$

where

$$\begin{aligned}\dot{y}_i =\ & x_i - \mathrm{Prox}_{C_{i2}}(x_i + \gamma_1 y_i) \\ \dot{z}_i =\ & \theta x_i - \theta\mathrm{Prox}_{\mathcal{D}_i}(x_i + \gamma_2 z_i) \\ \dot{\lambda}_i =\ & \theta\gamma_2(x_i - b_i) - \nabla C_{i1}(x_i) - \gamma_1 y_i - \theta\gamma_2 z_i - \theta\gamma_2\lambda_i \\ & - c\sum_{j=1}^{N} a_{ij}(\lambda_i - \lambda_j) - c\sum_{j=1}^{N} a_{ij}(w_i - w_j) \\ \dot{w}_i =\ & c\sum_{j=1}^{N} a_{ij}(\lambda_i - \lambda_j)\end{aligned}$$

and the other parameters are defined as the same as those in the neural network (8).

Denote $\varpi_i = \varepsilon_i - \breve{\varepsilon}_i$, $\varpi = \mathrm{col}(\varpi_i)$, $\varrho = \mathrm{diag}(\varrho_i) \otimes I_d$, and $\xi = \mathrm{col}(\xi_i)$, the closed-loop distributed neural network dynamics of (6) with (18) and (19) can be written as

$$\begin{aligned}\dot{x} &= -\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} + \varpi, \\ \dot{y} &= x - \mathrm{Prox}_{C_2}(x + \gamma_1 y) \\ \dot{z} &= \theta x - \theta\mathrm{Prox}_{\mathcal{D}}(x + \gamma_2 z) \\ \dot{\lambda} &= \theta\gamma_2(x - b) - \nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - cL\lambda - cLw \\ \dot{w} &= cL\lambda \\ \dot{\xi} &= -\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} - \varrho\operatorname{sign}(\varpi).\end{aligned} \tag{20}$$

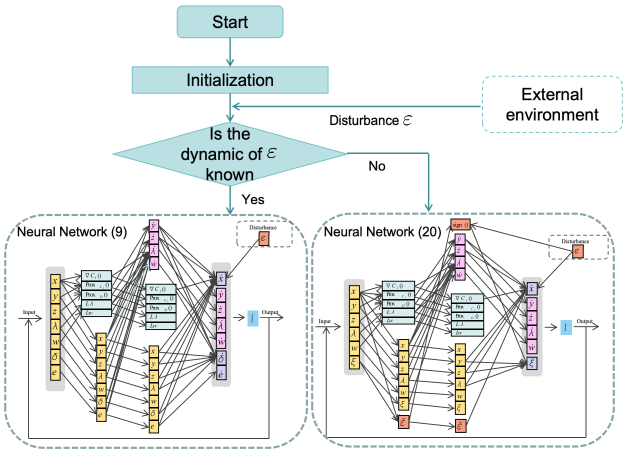

Fig. 1. Architecture of the proposed anti-disturbance neural networks.

In particular, the variable x represents the primal decision variables of the composite resource allocation problem (1), whose evolution encodes the distributed optimization process by integrating gradient information, proximal mappings, and disturbance compensation. The auxiliary variables y, z, λ, and w serve similar roles as in the structured case. In addition, the extra variable ξ compensates for unstructured disturbances by adaptively tracking their effects through discontinuous injection, thereby enabling suppression under bounded disturbance variation rates.

To provide a clear overview of the proposed approach, the main structure of this work is illustrated in Fig. 1. The flowchart at the top shows the network selection procedure based on whether the disturbance dynamics are known. The two neural network structures at the bottom correspond to the two proposed approaches for structured and unstructured disturbances, respectively. Each horizontal line between layers represents a linear combination of the required information for state updates. The nodes in the hidden layers correspond to the key computational modules, including gradient evaluation, proximal mappings, and disturbance compensation.

Remark 6: The bounded assumption of the perturbation rate of change ensures the controllability of disturbances. Compared with the literature [51], [52], [53], which only studies system models with structured disturbances (7), the more general disturbances discussed in this article make the proximal neural network (20) more applicable.

The use of the sign function in the proposed observer (18) may introduce non-Lipschitz behavior, which could lead to chattering phenomena. To address this open problem in numerical implementations, we believe that a feasible alternative is to use the saturation function (i.e., $\mathrm{sat}(x/\epsilon)$ with $\epsilon > 0$) instead of the sign function. By introducing a smooth transition near the origin, the chattering error caused by discontinuous switching can be effectively alleviated.

Theorem 3: Under Assumption 1 and proper parameters $\gamma_1 = \theta\gamma_2$, $\gamma_2 > l^2/\theta\sigma$, $c > ((3\theta\gamma_2\|L\|^2)/2 + (l\|L\|^2)/2)\max\{(1)/\chi_2(L), (1)/\chi_2^2(L)\}$, the state $x(t)$ generated by neural network (20) asymptotically converges to the optimal solution of composite resource allocation (1) under the multiagent system (6) with unstructured disturbances.

Proof: First, we prove that the equilibrium point $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, \xi^*)$ of neural network (20) implies that x^* is the optimal solution of resource allocation (2). According to the property of the equilibrium point, we have

$$\begin{aligned}
(0)_{Nd} &= -\nabla C_1(x^*) - \gamma_1 y^* - \theta\gamma_2 z^* - \theta\gamma_2\lambda^* + \varpi^*\\
x^* &= \mathrm{Prox}_{C_2}(x^* + \gamma_1 y^*)\\
x^* &= \mathrm{Prox}_{\mathcal{D}}(x^* + \gamma_2 z^*)\\
(0)_{Nd} &= \theta\gamma_2(x^* - b) - \varpi^* - cLw^*\\
(0)_{Nd} &= cL\lambda^*\\
\varpi^* &= \varrho\,\mathrm{sign}(\varpi^*).
\end{aligned}$$

Obviously, $\varpi^* = \varrho\,\mathrm{sign}(\varpi^*)$ if and only if $\varpi^* = (0)_{Nd}$. Thus, combining with the proof of Theorem 1, we have x^* is the optimal solution of resource allocation (2). Moreover, from Theorem 1, it also can be obtained that if x^* is the optimal solution of (2), there is $\mathrm{col}(x^*, y^*, z^*, \lambda^*, w^*, \varpi^*)$ to be the equilibrium point of neural network (9).

To proceed, we prove the state solution of the neural network (20) converges to the equilibrium point. Define the candidate function $W = V_1 + \phi_3/2\|\varpi\|^2$, where the form of V_1 can be found in Theorem 2, $\phi_1 < \min\{1, (\sigma/2 - l^2/(2\theta\gamma_2))/(\theta\gamma_2 + l)\}\phi_2$, $\phi_2 > 0$, and $\phi_3 > \phi_2/2\sigma$. Then similar to Theorem 2, the time derivative of W along the neural network (20) is

$$\begin{aligned}
\dot{W} = \;& \phi_2\bar{x}^\top\left[-\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z} + \varpi\right]\\
&+ \phi_2\gamma_1\bar{y}^\top\dot{y} + 2\phi_2\bar{y}^\top\dot{y} + \phi_2\bar{x}^\top\dot{y} + \phi_2\bar{y}^\top\dot{x} + \phi_2\gamma_2\bar{z}^\top\dot{z}\\
&+ 2\phi_2\bar{z}^\top\dot{z} + \phi_2\bar{x}^\top\dot{z} + \phi_2\bar{z}^\top\dot{x} + 2\phi_2\bar{y}^\top\dot{z} + 2\phi_2\bar{z}^\top\dot{y}\\
&+ \phi_2\bar{\lambda}^\top\left[\theta\gamma_2 x - \theta\gamma_2 b - \nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda\right]\\
&+ \phi_2\bar{\lambda}^\top\left[cL\lambda - cLw\right] + c\phi_2\bar{w}^\top L\lambda\\
&+ \phi_1\left(\bar{\lambda} + \bar{w}\right)^\top L\left[\theta\gamma_2 x - \theta\gamma_2 b - \nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z\right]\\
&+ \phi_1\left(\bar{\lambda} + \bar{w}\right)^\top L\left[-\theta\gamma_2\lambda - cLw\right] + c\phi_1\bar{w}^\top L^2\lambda\\
&+ \phi_3\sum_{i=1}^{N}\varpi_i^\top\left(\dot{\varepsilon}_i - \varpi_i - \varrho_i\mathrm{sign}(\varpi_i)\right)\\
\le & -\kappa_1\|\bar{x}\|^2 - \phi_2\|\dot{y}\|^2 - \frac{\phi_2}{\theta}\|\dot{z}\|^2 - \kappa_2\|\varpi\|^2\\
& -\kappa_3\left\|\bar{y} + \bar{z} + \bar{\lambda}\right\|^2 - \kappa_4\left\|\lambda^\perp\right\|^2 - \kappa_5\|\bar{w}\|^2
\end{aligned}\tag{21}$$

where $\kappa_1 = (\phi_2\sigma)/2 - (\phi_2 l^2)/(2\theta\gamma_2) - \phi_1\theta\gamma_2 - \phi_1 l$, $\kappa_2 = \phi_3 - \phi_2/2\sigma$, $\kappa_3 = (\phi_2\theta\gamma_2)/2 - (\phi_1\theta\gamma_2)/2$, $\kappa_4 = c\phi_2\chi_2(L) - (3\phi_1\theta\gamma_2\|L\|^2)/2 - (\phi_1 l\|L\|^2)/2$, $\kappa_5 = c\phi_1\chi_2^2(L) - (3\phi_1\theta\gamma_2\|L\|^2)/2 - (\phi_1 l\|L\|^2)/2$.

Therefore, based on LaSalle's invariance principle, if θ is big enough, and the parameters satisfy $\gamma_1 = \theta\gamma_2$, $\gamma_2 > l^2/\theta\sigma$, and $c > ((3\theta\gamma_2\|L\|^2)/2 + (l\|L\|^2)/2)\max\{1/(\chi_2(L)), 1/(\chi_2^2(L))\}$, the state $x(t)$ generated by the neural network (20) asymptotically converges to the optimal solution of composite resource allocation (1).

Remark 7: In the presence of unstructured external disturbances, observer-based compensation strategies play a key role in maintaining the accuracy of distributed neural networks. The term ξ_i in the observer (18) is designed to suppress disturbances by estimating the deviation between the state and its observations. In addition, the term $\text{sign}(\varepsilon-\tilde{\varepsilon}_i)$ contained in the observer (18) acts as a sliding mode correction, enhancing the robustness of the system under unstructured but bounded disturbances.

Notably, the disturbances embedded in the auxiliary layer of neural networks (9) and (20) can be obtained in real time by additional noise sensors, similar to the neural network design in [53], which will not affect the realizability of the proposed control protocols.

In practical applications, the proposed frameworks naturally address several real-world challenges. Hardware limitations are reflected in the local set constraints, computational demands are mitigated by avoiding multivalued subgradient operations, and sensor noise is treated as a form of external disturbance, which is explicitly handled by the designed neural networks. These correspondences demonstrate that the current framework not only guarantees theoretical convergence but also inherently accommodates key practical constraints and disturbances, underscoring its effectiveness and applicability in realistic distributed optimization scenarios.

Remark 8: Note that both proposed neural networks (9) and (20) are developed as improvements of a baseline distributed continuous-time neural dynamics formulated for the disturbance-free case. The baseline dynamics are given by

$$\begin{aligned}
\dot{x} &= -\nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - 2\dot{y} - 2\dot{z},\\
\dot{y} &= x - \text{Prox}C_2(x+\gamma_1 y)\\
\dot{z} &= \theta x - \theta\text{Prox}\mathcal{D}(x+\gamma_2 z)\\
\dot{\lambda} &= \theta\gamma_2(x-b) - \nabla C_1(x) - \gamma_1 y - \theta\gamma_2 z - \theta\gamma_2\lambda - cL\lambda - cLw\\
\dot{w} &= cL\lambda.
\end{aligned} \tag{22}$$

Although (22) is formulated in continuous time, for numerical implementation and practical applications, it can be discretized using an explicit Euler scheme with step size Δt

$$\begin{aligned}
x^{k+1} &= x^k + \Delta t\left[-\nabla C_1\left(x^k\right) - \gamma_1 y^k - \theta\gamma_2 z^k - \theta\gamma_2\lambda^k\right]\\
&\quad - 2\left(y^{k+1} - y^k\right) - 2\left(z^{k+1} - z^k\right)\\
y^{k+1} &= y^k + \Delta t\left[x^k - \text{Prox}C_2\left(x^k + \gamma_1 y^k\right)\right]\\
z^{k+1} &= z^k + \Delta t\left[\theta x^k - \theta\text{Prox}\mathcal{D}\left(x^k + \gamma_2 z^k\right)\right]\\
\lambda^{k+1} &= \lambda^k + \Delta t\left[\theta\gamma_2\left(x^k - b\right) - \nabla C_1\left(x^k\right) - \gamma_1 y^k - \theta\gamma_2 z^k\right.\\
&\quad \left. - \theta\gamma_2\lambda^k - cL\lambda^k - cLw^k\right]\\
w^{k+1} &= w^k + \Delta t cL\lambda^k.
\end{aligned}$$

At each iteration, the update rules involve three major computational components: 1) local gradient evaluations; 2) proximal operations; and 3) matrix–vector multiplications, so the per-step computational complexity of the baseline algorithm (22) is of polynomial order. Since the neural networks (9) and (20) retain the same basic update structure and only introduce additional disturbance-compensation terms, their computational complexity remains of the same polynomial order.

Furthermore, we emphasize that integrating proximal operators into the exact penalty framework does not significantly increase the computational complexity as existing distributed proximal neural networks [17], [18], [19], [20], [40]. This is because the additional proximal operator involved in our approach admits a closed-form solution for common distance functions, making its computational cost comparable to the standard proximal steps for the nonsmooth part of the cost as required in [17], [18], [19], [20], and [40]. This ensures that the proposed networks are scalable and can be efficiently applied to real-time, large-scale distributed systems.

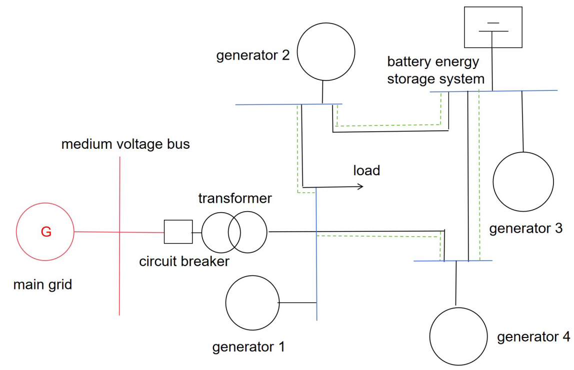

Fig. 2. Structure of power grid system.

V. SIMULATIONS

This section presents simulation results on resource allocation strategies within the power grid system under three distinct operational scenarios: normal operation without disturbances, operation with structured disturbances, and operation with unstructured disturbances.

Consider a power grid system with four generators. Fig. 2 shows the physical layout of the grid and the connection among the generators. According to [55], a typical power grid system consists of several key components, including the main grid, medium-voltage bus, circuit breakers, transformers, loads, and a battery energy storage system. In this system, generators play a critical role in regulating generation capacity to achieve an optimal power generation strategy, ensuring that the energy supply and demand remain balanced while optimizing efficiency and reliability.

In this example, let $x_i = (x_{i1}, x_{i2}) \in \mathbb{R}^2$ be the generation capacity of generator i, where x_{i1} and x_{i2} represent the power generation in the peak period and the trough period, respectively. The generation cost function of generator i is denoted by $C_i(x_i)$, its formulation and private information are listed in Table I. In addition, structured and unstructured disturbances from the external environment are considered, respectively.

In preparation for running distributed proximal neural networks designed in this article, the calculation procedure of proximal operators is briefly described below. First, for nonsmooth terms $C_{i2}(x_i) = \|x_i - o_i\|_1 (o_i \in \mathbb{R}^2)$ in local objective functions, let $v_i \in \mathbb{R}^2$ for $i \in \mathcal{I}_4$, it has

$$\text{Prox}_{C_{i2}}(v_i) = \begin{cases} v_i - \dfrac{v_i - o_i}{\|v_i - o_i\|} & \text{if } \|v_i - o_i\| > 1\\ o_i & \text{if } \|v_i - o_i\| \le 1. \end{cases}$$

Moreover, the proximal operators for local set constraints are calculated as follows:

$$\text{Prox}_{\mathcal{D}_i}(v_i) = \begin{cases} \mathcal{P}_{\Omega_i}(v_i) & \text{if } 0 < a < 1\\ \frac{1}{a}\mathcal{P}_{\Omega_i}(v_i) + \left(1 - \frac{1}{a}\right)v_i & \text{if } a \ge 1 \end{cases}$$

where $a_i = \|v_i - \mathcal{P}_{\Omega_i}(v_i)\|$.

TABLE I
INDIVIDUAL INFORMATION OF GENERATORS

Generator	**Cost function** $C_i(x_i)$	**Demand** b_i	**Feasible set** Ω_i
1	$\|x_1\|^2 + \|x_1\|_1$	$(2,3)^T$	$2 \leq \|x_1 - (2,2)^T\| \leq 3$
2	$\|x_2\|^2 + \dfrac{x_{21}^2}{100\sqrt{x_{21}^2+1}} + \dfrac{x_{22}^2}{100\sqrt{x_{22}^2+1}} + \|x_2\|_1$	$(0.5,1.5)^T$	$0 \leq x_{21} \leq 1,\quad 2 \leq x_{22} \leq 3,$ $\|x_2 - (1,2)^T\| \leq 1$
3	$\|x_3\|^2 + \|x_3\|_1$	$(4,3)^T$	$2 \leq \|x_3 - (4,3)^T\| \leq 3$
4	$\ln(\mathrm{e}^{-0.05x_{41}}+\mathrm{e}^{-0.05x_{41}})+\ln(\mathrm{e}^{-0.05x_{42}}+\mathrm{e}^{-0.05x_{42}})+$ $\|x_4\|^2 + \|x_4\|_1$	$(1.5,0.5)^T$	$1 \leq x_{41} \leq 2,\quad \|x_{42}\| \leq 1,$ $\|x_4 - (1,1)^T\| \leq 1$

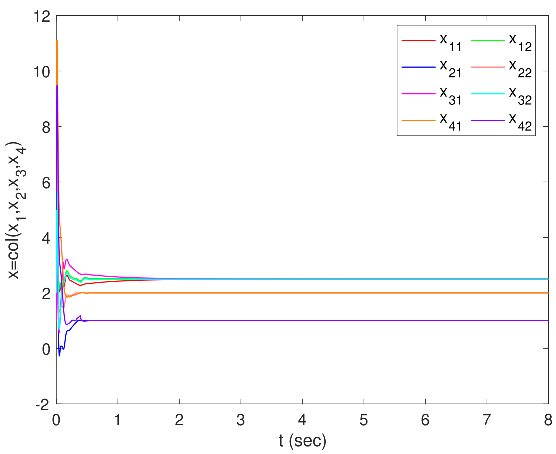

Fig. 3. Trajectories of x generated by the neural network (9) without disturbances.

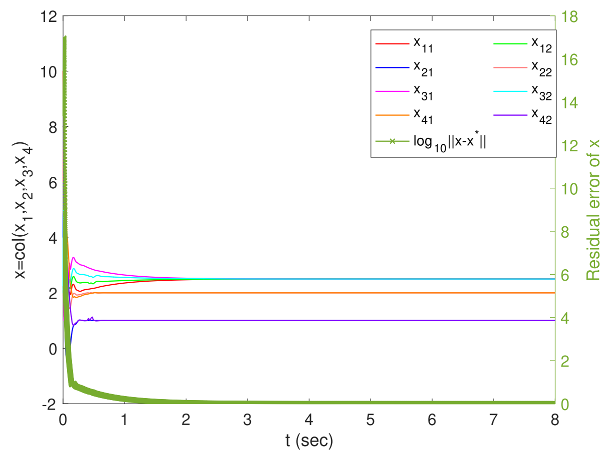

Fig. 5. Trajectories of x and $\|x - x^*\|$ generated by the neural network (20) with unstructured disturbances.

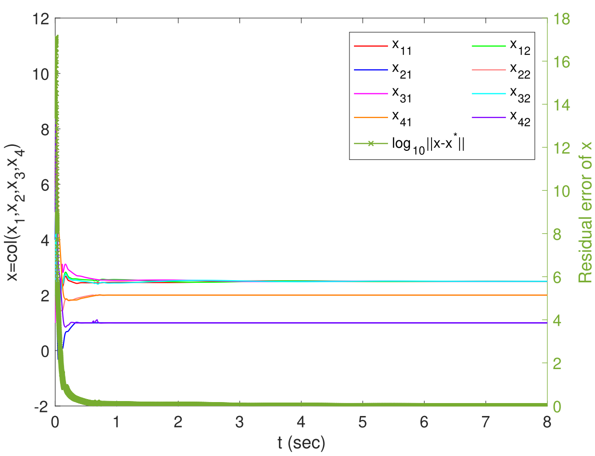

Fig. 4. Trajectories of x and $\|x - x^*\|$ generated by the neural network (9) with structured disturbances (7).

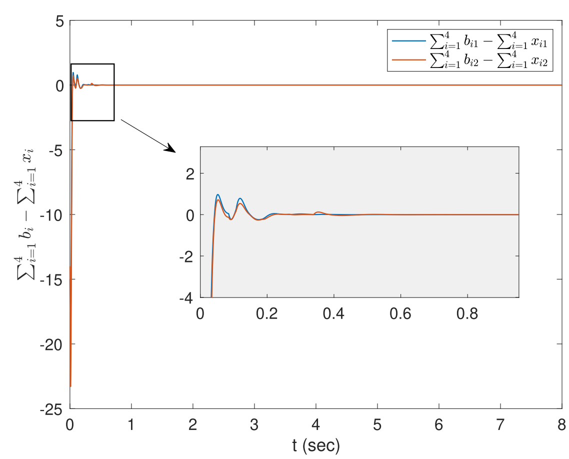

Fig. 6. Trajectories of coupling equality constraint.

For the structured disturbances (7) with $A_i = \begin{pmatrix} 0 & -i \\ -1 & 0 \end{pmatrix}$, $B_i = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$, $i = 1,2,3,4$. Set $K_i = \begin{pmatrix} 1 & -1 \\ 4 & -2 \end{pmatrix}$, $i = 1,2,3,4$, it holds that for each i, $A_i + K_iB_i$ is Hurwitz. By choosing parameters in neural network (9) as $\gamma_1 = 100$, $\gamma_2 = 2$, $\theta = 50$, and $c = 100$, the initial value of $x = \mathrm{col}(x_1, x_2, x_3, x_4)$ is set as $x(0) = (10,9,4,6,1,2,8,5)^\top$. In order to demonstrate the effectiveness of the proposed neural network in disturbance suppression, neural network (22) is first applied under disturbance-free conditions (i.e., $\varepsilon_i = 0$), and the neural network (9) is then tested under structured disturbances (7). The results are presented in Figs. 3 and 4, respectively. It is observed that the state solutions $x(t)$ obtained in both cases converge to $x^* = (2.5, 2.5, 1, 2, 2.5, 2.5, 2, 1)^\top$, which means that the neural network (9) is valid for the case of structured disturbances.

Following the above parameters γ_1, γ_2, θ, and c, set $\xi(0) = (1)_8$, then under the unstructured disturbances $\varepsilon_i = (i\sin(t), i\cos(t))$, the trajectories of $x(t)$ generated by neural network (20) is given in Fig. 5. In Fig. 6, it shows that the obtained optimal allocation strategy satisfies the constraint of

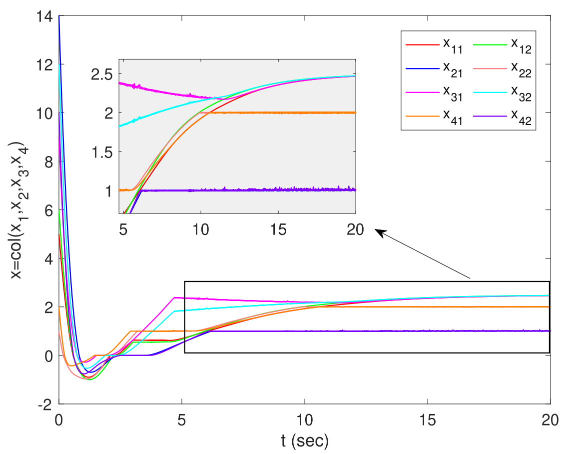

Fig. 7. Trajectories of x generated by the subgradient-based method in [24].

energy supply and demand balance. To sum up, the proximal neural networks proposed in this article can effectively obtain the optimal generation strategy of the above power grid system under structured and unstructured disturbances.

To provide a fair performance evaluation, we compare the proposed algorithm with the subgradient-based distributed method in [24]. As shown in Fig. 7, both methods achieve similar steady-state accuracy. However, the algorithm in [24] converges more slowly and generates oscillatory trajectories, which can be attributed to the inherent multivalued nature of subgradient dynamics when the optimal solution lies on the boundary of local constraint sets. In contrast, the proposed neural networks employ proximal operators that smooth the update process, accelerate convergence, and yield stable trajectories. These properties make it more suitable for high-accuracy engineering applications.

VI. Conclusion

This article addressed nonsmooth composite resource allocation problems with coupled and local set constraints in multiagent systems under external disturbances. To avoid multivalued subgradients, two proximal neural networks were proposed: one based on the internal model principle to suppress structured disturbances, and an observer-based variant to enhance robustness against unstructured disturbances. Both networks were theoretically proven to converge to the exact optimal solution, with numerical simulations confirming their effectiveness. Overall, these results demonstrate the proposed frameworks' capability to achieve accurate, efficient, and robust solutions for composite resource allocations.

Future work will build upon the current study and focus on extending the proposed frameworks to more complex and realistic scenarios, including time-varying network topologies, asynchronous communications, and event-triggered implementations aimed at further enhancing communication efficiency.

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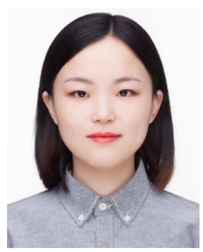

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